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Data Analytics

UNION and UNION ALL Exercises

Question 1: UNION

```
SELECT account_id, account_holder, city
FROM branch_sandton_account
UNION
```

```
SELECT account_id, account_holder, city
FROM branch_rosebank_accounts;
```

account_id	account_holder	city
A001	Namvula Dlamini	Johannesburg
A002	David Mokaena	Johannesburg
A003	Lerato Sithole	Johannesburg
A004	Peter Nkosi	Johannesburg
A005	Zande Khumalo	Johannesburg
A006	Thabo Motsha	Johannesburg

Question 2: UNION

```
SELECT product_code, product_name, product_type
FROM savings_products
UNION
```

```
SELECT product_code, product_name, product_type
FROM current_products;
```

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Product_code	Product_name	Product_type
Sv01	Basic Savings	Savings
Sv02	Premium Savings	Savings
Sv03	Youth Savings	Savings
Sv04	Business Savings	Savings
Cr01	Standard Current	Current
Cr02	Gold Current	Current
Cr03	Business Current	Current

Question 3

```
SELECT staff_id, staff_name, email
FROM retail_banking_staff
```

UNION

```
SELECT staff_id, staff_name, email
FROM corporate_banking_staff;
```

staff_id	staff_name	email
5001	Mpho Radeke	mpho.radeke@redbank.co.za
5002	Brian Tshabalala	brian.tshabalala@redbank.co.za
5003	Aisha Patel	aisha.patel@redbank.co.za
5004	Kabalo Mbalolo	kabalo.mbalolo@redbank.co.za

Question 4

```
SELECT city_code, city_name, region
FROM mobile_branch_cities
```

UNION

```
SELECT city_code, city_name, region
FROM digital_branch_cities;
```

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city_code	city_name	region
ca	Johannesburg	Gauteng
ce	Pretoria	Gauteng
cs	Cape Town	Western Cape
cz	Durban	KwaZulu - Natal
cs	Polokwane	Limpopo
ce	Port Elizabeth	Eastern Cape

Question 5

```
SELECT customer_id, customer_name, segment
FROM push_notification_targets
UNION
```

```
SELECT customer_id, customer_name, segment
FROM inapp_banner_targets;
```

customer_id	customer_name	segment
c1001	Nomsa Zwane	Retail
c1002	Andile Buthelez	Premium
c1003	Fahima Mohamed	Retail
c1004	Ryno van Zyl	Premium
c1005	Thandeka Cele	Retail
c1006	Samuel Nkosi	Premium

Question 6

```
SELECT txn_id AS transaction_id, account_id, amount,
transaction_date
FROM atm01_transactions
UNION ALL
```

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```
SELECT txn_id AS transaction_id, account_id, amount,
transaction date
FROM atm02_transactions;
```

transaction_id	account_id	amount	transactiondate
T1001	A001	500.00	2026-03-02
T1002	A002	1200.00	2026-03-03
T1003	A001	300.00	2026-03-04
T1004	A003	750.00	2026-03-05
T1005	A001	300.00	2026-03-04
T1006	A004	900.00	2026-03-06
T1007	A002	450.00	2026-03-07
T1008	A005	1500.00	2026-03-08

Question 7

```
SELECT app_id AS application_id, customer_id,
loan_type, amount_requested
FROM gauteng_loan_applications
UNION ALL
```

```
SELECT app_id AS application_id, customer_id,
loan_type, amount_requested
FROM western_cape_loan_applications;
```

application_id	customer_id	loan_type	amount_requested
LA001	C1001	Home loan	850000.00
LA002	C1002	Vehicle loan	220000.00
LA003	C1003	Personal loan	60000.00
LA004	C1004	Home loan	620000.00
LA005	C1005	Personal loan	60000.00

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application_id	customer_id	loan type	amount requested
LA005	C1005	Vehicle loan	180 000 £
LA006	C1006	Business	500 000
LA007	C007	Home loan	900 000 £

Question 8

```
SELECT complaint_id, customer_id, category, logged_date
FROM email_complaints
```

UNION ALL

```
SELECT complaint_id, customer_id, category, logged_date
FROM app_complaints;
```

complaint_id	customer_id	category	logged_date
EC001	C2001	Fees	2026-01-01
EC002	C2002	App Issue	2026-02-03
EC003	C2003	Card Issue	2026-04-05
EC004	C2004	Service	2026-04-07
EC005 AC001	C2005	Service	2026-04-08
AC002	C2001	Fees	2026-04-04
AC003	C2006	App Issue	2026-04-06
AC004	C2007	Card Issue	2026-04-08

Question 9

```
SELECT payment_id, account_id, amount, payment_date
FROM april_payments
```

UNION ALL

```
SELECT payment_id, account_id, amount, payment_date
FROM may_payments;
```

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payment_id	account_id	amount	payment-date
PA4001	AC01	12500.00	2026-04-05
PA4002	AC02	4800.00	2026-04-10
PA4003	AC03	9200.00	2026-04-15
PA4004	AC04	3300.00	2026-4-20
PA4005	AC01	12500.00	2026-05-05
PA4006	AC05	7600.00	2026-05-10
PA4007	AC02	5700.00	2026-05-15
PA4008	AC06	2800.00	2026-05-20

Question 10

```
SELECT entry_id, account_id, entry_type, amount,
       entry_date
FROM   debit_entries
UNION ALL
```

```
SELECT entry_id, account_id, entry_type, amount,
       entry_date
FROM   credit_entries;
```

entry_id	account_id	entry_type	amount	entry_date
DR001	AC01	Debit	1500.00	2026-06-01
DR002	AC02	Debit	2200.00	2026-06-02
DR003	AC03	Debit	800.00	2026-06-03
DR004	AC04	Debit	3100.00	2026-06-04
CR001	AC01	Credit	1500.00	2026-06-01
CR002	AC05	Credit	4200.00	2026-06-02
CR003	AC02	Credit	2200.00	2026-06-03
CR004	AC06	Credit	950.00	2026-06-04

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Bonus Question 1

They should use Union because the goal is to create a list of unique customers. If the same customer appears on both the January and February watch lists, they should only be included once. Union automatically removes duplicate records, making it the right choice for producing a distinct list of high risk customers. If UNION ALL were used then customers who appear in both months would be listed twice. This would result in an inflated watch list and an inaccurate count of the actual number of high risk individuals.

Bonus Question 2

UNION ALL needs to be used as the auditing team needs every transaction to be included in the results, even if the same transaction appears in both systems on the same day. Using UNION would remove duplicate records, which could lead to transactions being excluded and the total transaction count being lower than it should be. UNION ALL keeps every record from both tables, making it the better choice when accurate transaction counts are essential for tasks such as auditing, reconciliations, and verifying ledger totals.

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Bonus Question 3

The query doesn't work because both `SELECT` statements must return the same number of columns. In the case, the first `SELECT` returns only two columns `account_id` and `account_holder` while the second returns three columns `account_id`, `account_holder`, and `city`. Since `UNION` requires both queries to have the same number of columns in the same order, the database returns a column mismatch error.

To fix this, make sure both `SELECT` statements return the same number of columns. If the `city` column exists in the `branch_sandton_accounts` table, include it in the first query as well. If it doesn't, remove the `city` column from the second query so that both queries return only `account_id` and `account_holder`. This will align the queries and produce the expected results for Question 1.

```
SELECT account_id, account_holder, city  
FROM branch_sandton_accounts  
UNION
```

```
SELECT account_id, account_holder, city  
FROM branch_rosebank_accounts;
```